Graded exercises

E1 · Direct application. In an orthonormal basis of space, consider $\vec{u} = (2;-1;3)$ and $\vec{v} = (1;4;0)$. Calculate $\vec{u} \cdot \vec{v}$.

E2 · Application and justification. In an orthonormal basis, consider $\vec{u} = (a;2;1)$ and $\vec{v} = (3;-1;4)$ where a is real. Determine a so that $\vec{u}$ and $\vec{v}$ are orthogonal.

E3 · Method to organise. In an orthonormal coordinate system, a plane is spanned by $\vec{u} = (1;-1;0)$ and $\vec{v} = (1;0;-1)$. Show that $\vec{n} = (1;1;1)$ is normal to it. We then consider $A(0;0;0)$ and $B(2;2;2)$. Describe geometrically the set of points equidistant from A and B.

E4 · Choice of strategy. In an orthonormal basis, consider $\vec{u} = (1;0;1)$ and $\vec{v} = (0;1;1)$. Determine the measure, between 0 and π, of the angle between these two vectors.

E5 · Synthesis or proof. A tetrahedron has as its base a right-angled triangle whose two perpendicular sides measure 3 cm and 4 cm. The height of the tetrahedron relative to this base is 5 cm. Calculate its volume in cm ^{3}.